Assignment-1

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1) Breast Cancer Classification Performance.

→ Find the Missing False Negatives

- Dataset per class = 1800
- Test data = 20%
- Test samples per class = 20% of 1800 = 360

For the Malignant (positive) class: TP + FN = 360

Given:

- TP = 210

Therefore, FN = 360 − 210 = 150

False Negatives (FN) = 150

→ calculate Performance Metrics.

Given:

- TP = 210
- TN = 190
- FP = 30
- FN = 150

1. Accuracy

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

$$= \frac{210+190}{210+190+30+150} = \frac{400}{580} = 0.6897$$

Accuracy = 68.97%

2. precision

$$\text{Precision} = \frac{TP}{TP+FP}$$

$$= \frac{210}{210+30} = \frac{210}{240} = 0.875$$

precision = 87.5%

3. Sensitivity (Recall)

$$\text{Sensitivity} = \frac{TP}{TP+FN}$$

$$= \frac{210}{210+150} = \frac{210}{360} = 0.5833$$

sensitivity = 58.33%

4. specificity

$$\text{specificity} = \frac{TN}{TN+FP}$$

$$= \frac{190}{190+30} = \frac{190}{220} = 0.8636$$

Specificity = 86.36%

Final Results :-

Metric	value
False Negatives (FN)	150
Accuracy	68.97%
precision	87.50%
Sensitivity	58.33%
Specificity	86.36%

Effectiveness in Medical Diagnosis:-

- The Model has high precision (87.5 %), indicating that when it predicts cancer, it is usually correct.
- The specificity (86.36 %) is also high, meaning it correctly Identifies most benign cases.
- However, the Sensitivity (58.33 %) is relatively low, Indicating that many malignant cases are missed. (high FN = 150), which is undesirable in medical because missed cancer cases can delay diagnosis Treatment
- Therefore, Improving sensitivity should be the primary objective for this Classifier.

2) Model performance Analysis

Given Data

Epoch	Training loss	Validation loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

i) Epoch at which Overfitting Begins.

up to Epoch 3, both Training and validation losses decrease.

From Epoch 4 onwards.

- Training loss continues to decrease.
- Validation loss starts increasing.

→ overfitting begins at Epoch 4.

ii) Two Techniques to Reduce overfitting

1. Early stopping
   - Stop Training when the validation loss starts increasing
2. Dropout Regularization.
   - Randomly deactivate neurons during Training to prevent the model from memorizing the Training data.

iii) Effect of overfitting on Generalization:-

- overfitting causes the model to learn noise and specific patterns from the training data Instead of general features.
- As a Result, it performs very well on Training data but poorly on unseen data.
- This Reduces the Model's generalization ability, leading to lower prediction accuracy in real-world applications.